

20251006EB_MN (vLH51)

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2025年10月6日 星期一

split iPSCs into a full 6-well plate on the Friday before differentiation
differentiate 5 of 6-wells with 10mL T2 in cell-repellent plates
keep some iPSCs

2025年10月8日 星期三

T2 -> T3

2025年10月10日 星期五

T3 -> T4 (~14 mL for over-weekend)

2025年10月13日 星期一

T4 -> T5

2025年10月15日 星期三

T5 -> T6

2025年10月17日 星期五

T6 -> T6 (~14mL for over-weekend)

2025年10月20日 星期一

T6 -> T7

Dissociation

1. Collect EBs in 50mL falcon and let the EB settle with gravity
2. Wash EBs with 3mL PBS
3. Spin EBs down with 100rcf, 1min. Remove the supernatant (can leave some liquid to avoid remove EBs).
4. Add 3mL 0.025% trypsin (diluted from 0.25% with DNase) and pipette it up and down with 5mL serological pipette.
5. Put the tube into 37C water bath.
6. Pipette the cells with P1000 at 5 min and flick at ~10min.
7. Pipette and collect the sup into 6mL HI-FBS.
8. Add 3mL 0.025% trypsin with DNase.
9. Pipette at 21min, flick at 25min.
10. Pipette at 32min and transfered all to the same 6mL HI-FBS.
11. Add 6mL of CTWM and pipette it with 5mL serological pipette gently.

12. accidentally centrifuge at 200xg, 2.5min without filtering.
13. Resuspend the cells with 10mL pipette.
14. Filter the cell with 40um filter into a new 50mL tube
15. Spin the cell down at 200rcf, 3.5min. Remove the supernatent.
16. Resuspend cells wtih 1mL T7.
17. Add 4mL more T7 to resuspend the cells and count it.
18. Plate the cells with T7. (350-500K/well for 12well plate; 1-1.5M/well for 6-well plate)

cell info				^
	A	B	C	
1	cell	passage	well#	
2	WT11	P8	5	

Table1			^
	A	B	
1	run1	volume (mL)	
2	trypsin (0.025%)	3	
3	HI-FBS/CTWM	6	
4	run2	*take some undissolved EB	
5	trypsin (0.025%)	3	
6	HI-FBS/CTWM	6	

Table3										/
	A	B	C	D	E	F	G	H	I	
1	conc. (M/mL)	survival	total volume	tota cell count	plate#	well plate	cell count/well (K)	number of wells	volume	
2	183	99	5	915	A	24	228.75	4	1.3	

Well1 ^

	1	2	3	4	5	6
A						
B			vLH5			
C			1. 2uL			
D			in total			

2025年10月21日 星期二

Change T7 -> T7

2025年10月23日 星期四

T7 -> T8 (all T8 below are BPT8)

2025年10月24日 星期五

1/2 T8 -> 1/2 T8

2025年10月27日 星期一

1/2 T8 -> 1/2 T8

2025年10月29日 星期三

1/2 T8 -> 1/2 T8

2025年10月30日 星期四

1/2 T8 -> 1/2 for A3 and B3
treat C3 amd D3 at 15:30 for 24h treatment

2025年10月31日 星期五

Imaging vLH51
microscope: old spinning disk confocal
60x, 1.5xzoom
0.1342um/pixel
for 1h movies: ROI = 600*900
for 24h movies: ROI = 650*1040
Note:
1. 24h movies are all more than 10.5-10.77ms/frame --> I didn't notice the default ROI of old spinning disk confocal is bigger. the ROI is bigger: 650*1040
2. not many GEM+ cells. Also, many GEM+ cells have too many GEMs
3. hard to find GEMs in neurites.

Table2 ^

	A	B	C	D
1	condition		51	
2				image
3	1	2.5mM 1h	12:00	13:20
4	2	0.15mM 1h	12:30	13:50
5	3	2.5mM 24h	10/30 15:30	14:50
6	4	0.15mM 24h	10/30 15:30	15:20

Well2 ^

	1	2	3	4	5	6
A			1			
B			2			
C			3			
D			4			

vLH51 ^

	A	B	C	D
1		laser power	exposure time	
2	488 10ms	100	10ms	movie

vLH51_fileid-conditions				
	A	B	C	D
1	image	treatment	treat_time	type
2	soma_vLH51_MN_60x_1.5xzoom_001	2.5mM gluc	1h	movie
3	soma_vLH51_MN_60x_1.5xzoom_002	2.5mM gluc	1h	movie
4	soma_vLH51_MN_60x_1.5xzoom_003	2.5mM gluc	1h	movie
5	soma_vLH51_MN_60x_1.5xzoom_004	2.5mM gluc	1h	movie
6	soma_vLH51_MN_60x_1.5xzoom_005	2.5mM gluc	1h	movie
7	soma_vLH51_MN_60x_1.5xzoom_006	2.5mM gluc	1h	movie
8	soma_vLH51_MN_60x_1.5xzoom_007	2.5mM gluc	1h	movie
9	soma_vLH51_MN_60x_1.5xzoom_008	2.5mM gluc	1h	movie
10	soma_vLH51_MN_60x_1.5xzoom_009	2.5mM gluc	1h	movie
11	soma_vLH51_MN_60x_1.5xzoom_010	2.5mM gluc	1h	movie
12	soma_vLH51_MN_60x_1.5xzoom_011	2.5mM gluc	1h	movie
13	soma_vLH51_MN_60x_1.5xzoom_012	2.5mM gluc	1h	movie
14	soma_vLH51_MN_60x_1.5xzoom_013	2.5mM gluc	1h	movie
15	soma_vLH51_MN_60x_1.5xzoom_014	2.5mM gluc	1h	movie
16	soma_vLH51_MN_60x_1.5xzoom_015	0.15mM gluc	1h	movie
17	soma_vLH51_MN_60x_1.5xzoom_016	0.15mM gluc	1h	movie
18	soma_vLH51_MN_60x_1.5xzoom_017	0.15mM gluc	1h	movie
19	soma_vLH51_MN_60x_1.5xzoom_018	0.15mM gluc	1h	movie
20	soma_vLH51_MN_60x_1.5xzoom_019	0.15mM gluc	1h	movie
21	soma_vLH51_MN_60x_1.5xzoom_020	0.15mM gluc	1h	movie
22	soma_vLH51_MN_60x_1.5xzoom_021	0.15mM gluc	1h	movie
23	soma_vLH51_MN_60x_1.5xzoom_022	0.15mM gluc	1h	movie
24	soma_vLH51_MN_60x_1.5xzoom_023	0.15mM gluc	1h	movie
25	soma_vLH51_MN_60x_1.5xzoom_024	0.15mM gluc	1h	movie
26	soma_vLH51_MN_60x_1.5xzoom_025	0.15mM gluc	1h	movie
27	soma_vLH51_MN_60x_1.5xzoom_026	0.15mM gluc	1h	movie
28	soma_vLH51_MN_60x_1.5xzoom_027	0.15mM gluc	1h	movie
29	soma_vLH51_MN_60x_1.5xzoom_028	0.15mM gluc	1h	movie
30	soma_vLH51_MN_60x_1.5xzoom_029	0.15mM gluc	1h	movie
31	soma_vLH51_MN_60x_1.5xzoom_030	0.15mM gluc	1h	movie
32	soma_vLH51_MN_60x_1.5xzoom_031	2.5mM gluc	24h	movie
33	soma_vLH51_MN_60x_1.5xzoom_032	2.5mM gluc	24h	movie

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34	soma_vLH51_MN_60x_1.5xzoom_033	2.5mM gluc	24h	movie
35	soma_vLH51_MN_60x_1.5xzoom_034	2.5mM gluc	24h	movie
36	soma_vLH51_MN_60x_1.5xzoom_035	2.5mM gluc	24h	movie
37	soma_vLH51_MN_60x_1.5xzoom_036	2.5mM gluc	24h	movie
38	soma_vLH51_MN_60x_1.5xzoom_037	2.5mM gluc	24h	movie
39	soma_vLH51_MN_60x_1.5xzoom_038	2.5mM gluc	24h	movie
40	soma_vLH51_MN_60x_1.5xzoom_039	2.5mM gluc	24h	movie
41	soma_vLH51_MN_60x_1.5xzoom_040	2.5mM gluc	24h	movie
42	soma_vLH51_MN_60x_1.5xzoom_041	2.5mM gluc	24h	movie
43	soma_vLH51_MN_60x_1.5xzoom_042	2.5mM gluc	24h	movie
44	soma_vLH51_MN_60x_1.5xzoom_043	2.5mM gluc	24h	movie
45	soma_vLH51_MN_60x_1.5xzoom_044	2.5mM gluc	24h	movie
46	soma_vLH51_MN_60x_1.5xzoom_045	2.5mM gluc	24h	movie
47	soma_vLH51_MN_60x_1.5xzoom_046	0.15mM gluc	24h	movie
48	soma_vLH51_MN_60x_1.5xzoom_047	0.15mM gluc	24h	movie
49	soma_vLH51_MN_60x_1.5xzoom_048	0.15mM gluc	24h	movie
50	soma_vLH51_MN_60x_1.5xzoom_049	0.15mM gluc	24h	movie
51	soma_vLH51_MN_60x_1.5xzoom_050	0.15mM gluc	24h	movie
52	soma_vLH51_MN_60x_1.5xzoom_051	0.15mM gluc	24h	movie
53	soma_vLH51_MN_60x_1.5xzoom_052	0.15mM gluc	24h	movie
54	soma_vLH51_MN_60x_1.5xzoom_053	0.15mM gluc	24h	movie
55	soma_vLH51_MN_60x_1.5xzoom_054	0.15mM gluc	24h	movie
56	soma_vLH51_MN_60x_1.5xzoom_055	0.15mM gluc	24h	movie
57	soma_vLH51_MN_60x_1.5xzoom_056	0.15mM gluc	24h	movie
58	soma_vLH51_MN_60x_1.5xzoom_057	0.15mM gluc	24h	movie
59	soma_vLH51_MN_60x_1.5xzoom_058	0.15mM gluc	24h	movie
60	soma_vLH51_MN_60x_1.5xzoom_059	0.15mM gluc	24h	movie
61	soma_vLH51_MN_60x_1.5xzoom_060	0.15mM gluc	24h	movie
62	soma_vLH51_MN_60x_1.5xzoom_061	0.15mM gluc	24h	movie
63	neurite_vLH51_MN_60x_1.5xzoom_001	2.5mM gluc	1h	movie
64	neurite_vLH51_MN_60x_1.5xzoom_002	2.5mM gluc	1h	movie
65	neurite_vLH51_MN_60x_1.5xzoom_003	2.5mM gluc	1h	movie
66	neurite_vLH51_MN_60x_1.5xzoom_004	2.5mM gluc	1h	movie
67	neurite_vLH51_MN_60x_1.5xzoom_005	2.5mM gluc	1h	movie
68	neurite_vLH51_MN_60x_1.5xzoom_006	2.5mM gluc	1h	movie
69	neurite_vLH51_MN_60x_1.5xzoom_007	2.5mM gluc	1h	movie
70	neurite_vLH51_MN_60x_1.5xzoom_008	2.5mM gluc	1h	movie

71	neurite_vLH51_MN_60x_1.5xzoom_009	2.5mM gluc	1h	movie
72	neurite_vLH51_MN_60x_1.5xzoom_010	2.5mM gluc	1h	movie
73	neurite_vLH51_MN_60x_1.5xzoom_011	0.15mM gluc	1h	movie
74	neurite_vLH51_MN_60x_1.5xzoom_012	0.15mM gluc	1h	movie
75	neurite_vLH51_MN_60x_1.5xzoom_013	0.15mM gluc	1h	movie
76	neurite_vLH51_MN_60x_1.5xzoom_014	0.15mM gluc	1h	movie
77	neurite_vLH51_MN_60x_1.5xzoom_015	0.15mM gluc	1h	movie
78	neurite_vLH51_MN_60x_1.5xzoom_016	0.15mM gluc	1h	movie
79	neurite_vLH51_MN_60x_1.5xzoom_017	0.15mM gluc	1h	movie
80	neurite_vLH51_MN_60x_1.5xzoom_018	0.15mM gluc	1h	movie
81	neurite_vLH51_MN_60x_1.5xzoom_019	0.15mM gluc	1h	movie
82	neurite_vLH51_MN_60x_1.5xzoom_020	0.15mM gluc	1h	movie
83	neurite_vLH51_MN_60x_1.5xzoom_021	2.5mM gluc	24h	movie
84	neurite_vLH51_MN_60x_1.5xzoom_022	2.5mM gluc	24h	movie
85	neurite_vLH51_MN_60x_1.5xzoom_023	2.5mM gluc	24h	movie
86	neurite_vLH51_MN_60x_1.5xzoom_024	2.5mM gluc	24h	movie
87	neurite_vLH51_MN_60x_1.5xzoom_025	2.5mM gluc	24h	movie
88	neurite_vLH51_MN_60x_1.5xzoom_026	2.5mM gluc	24h	movie
89	neurite_vLH51_MN_60x_1.5xzoom_027	2.5mM gluc	24h	movie
90	neurite_vLH51_MN_60x_1.5xzoom_028	2.5mM gluc	24h	movie
91	neurite_vLH51_MN_60x_1.5xzoom_029	2.5mM gluc	24h	movie
92	neurite_vLH51_MN_60x_1.5xzoom_030	2.5mM gluc	24h	movie
93	neurite_vLH51_MN_60x_1.5xzoom_031	0.15mM gluc	24h	movie
94	neurite_vLH51_MN_60x_1.5xzoom_032	0.15mM gluc	24h	movie
95	neurite_vLH51_MN_60x_1.5xzoom_033	0.15mM gluc	24h	movie
96	neurite_vLH51_MN_60x_1.5xzoom_034	0.15mM gluc	24h	movie
97	neurite_vLH51_MN_60x_1.5xzoom_035	0.15mM gluc	24h	movie
98	neurite_vLH51_MN_60x_1.5xzoom_036	0.15mM gluc	24h	movie
99	neurite_vLH51_MN_60x_1.5xzoom_037	0.15mM gluc	24h	movie
100	neurite_vLH51_MN_60x_1.5xzoom_038	0.15mM gluc	24h	movie
101	neurite_vLH51_MN_60x_1.5xzoom_039	0.15mM gluc	24h	movie
102	neurite_vLH51_MN_60x_1.5xzoom_040	0.15mM gluc	24h	movie